

NAJIRAN, Saudi Arabia

WMO: 411280

Lat: 17.611N

Lon: 44.419E

Elev: 3982

StdP: 12.70

Time Zone: 3.00 (E03)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	45.0	48.2	10.1	10.7	64.6	13.9	12.9	67.1	13.9	79.2	12.1	75.8	0.6	240	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	104.3	63.2	103.6	63.1	102.2	62.9	69.8	91.8	68.2	91.0	67.0	90.5	8.3	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.1	103.9	76.3	61.2	93.7	76.2	59.2	87.2	77.8	36.4	91.2	35.0	91.7	33.9	88.9	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.5	12.3	11.3	40.3	106.8	3.1	1.4	38.1	107.8	36.3	108.6	34.5	109.4	32.3	110.3		
			31.7	71.6	3.3	2.9	29.3	73.6	27.3	75.3	25.5	76.9	23.1	79.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.4	64.4	69.5	75.4	80.4	85.5	88.6	91.5	90.4	84.9	76.0	69.6	64.6
	DBStd	10.33	5.59	6.01	5.51	4.17	3.13	2.63	2.44	2.53	3.47	3.52	3.77	4.84
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	181	82	24	3	0	0	0	0	0	0	0	6	66
	CDD50	10383	446	547	788	911	1099	1157	1286	1252	1047	807	589	454
	CDD65	5091	63	152	326	462	634	707	821	787	597	342	146	54
	CDH74	75320	942	1928	3805	5844	9438	11445	13548	12756	8861	4330	1632	791
	CDH80	45371	297	822	1824	3172	5833	7582	9197	8507	5451	2033	457	196
Wind	WSAvg	4.6	3.6	4.2	5.0	5.0	4.9	5.1	6.2	5.9	4.8	3.9	3.4	3.2
Precipitation	PrecAvg	2.8	0.2	0.0	0.6	0.7	0.3	0.2	0.1	0.4	0.0	0.2	0.1	0.0
	PrecMax	10.4	1.9	0.2	6.4	3.6	2.1	3.0	1.0	2.5	0.1	3.3	1.1	0.2
	PrecMin	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.0	0.5	0.0	1.3	0.9	0.6	0.6	0.2	0.6	0.0	0.7	0.2	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.0	95.2	98.7	99.7	102.6	105.8	106.0	105.8	103.9	95.3	89.6	88.9
		MCWB	59.1	59.5	62.5	61.8	62.0	62.5	63.6	64.1	62.8	59.8	58.9	60.6
	2%	DB	87.4	91.7	95.4	97.2	100.8	104.0	104.3	104.0	100.8	93.2	87.3	84.8
		MCWB	58.5	59.5	61.1	61.4	61.9	62.4	63.3	63.8	62.2	59.2	59.1	59.1
	5%	DB	82.8	89.2	92.9	95.4	100.1	102.4	103.6	102.5	99.0	91.3	84.6	82.0
		MCWB	57.6	59.1	60.8	61.4	61.9	62.1	63.2	63.8	61.8	58.9	58.1	57.4
	10%	DB	79.2	85.7	89.6	93.5	98.4	100.8	102.1	101.3	97.2	89.5	82.5	78.6
		MCWB	56.4	58.3	60.2	60.9	61.6	61.6	63.1	63.8	61.4	58.6	57.3	56.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.3	66.4	68.8	69.8	70.8	71.1	71.1	71.2	67.7	67.8	65.2	64.3
		MCDB	81.0	83.2	87.4	87.8	94.8	92.5	95.3	92.1	93.5	77.3	81.1	82.2
	2%	WB	61.9	63.1	66.3	67.4	67.7	68.1	68.5	69.1	65.8	65.3	62.9	61.2
		MCDB	78.8	83.8	82.0	84.3	89.2	95.1	93.9	92.2	94.4	80.3	78.9	79.6
	5%	WB	59.6	61.1	64.5	66.0	66.1	66.0	67.0	67.6	64.1	63.1	61.2	59.4
		MCDB	77.4	83.1	82.5	83.7	88.2	94.0	94.0	93.3	94.1	83.1	77.6	77.7
	10%	WB	57.6	59.3	63.0	64.6	64.6	64.0	65.7	66.4	62.7	60.7	59.4	57.6
		MCDB	75.8	81.8	82.2	83.1	88.0	94.7	94.4	93.9	94.5	85.5	77.5	75.3
Mean Daily Temperature Range	5% DB	MDBR	27.1	27.8	26.0	25.1	25.6	26.2	22.3	23.5	26.1	27.0	25.6	26.6
		MCDBR	32.7	32.8	29.9	28.1	27.6	27.0	23.8	24.4	26.5	27.8	28.1	30.4
		MCWBR	14.7	13.8	11.8	10.4	10.0	10.0	8.6	8.8	10.2	11.5	12.1	13.6
	5% WB	MCDBR	27.6	28.1	24.1	23.3	22.9	24.4	21.1	22.6	25.1	22.8	23.0	26.0
		MCWBR	13.2	12.8	10.1	9.0	8.8	9.9	8.2	8.8	10.7	10.5	10.5	12.1
Clear-Sky Solar Irradiance	taub	0.390	0.431	0.493	0.511	0.549	0.614	0.669	0.637	0.530	0.409	0.393	0.374	
	taud	2.161	2.020	1.859	1.846	1.741	1.588	1.539	1.592	1.768	2.075	2.134	2.190	
	Ebn at Noon	282	275	262	257	245	226	215	224	248	279	278	283	
	Edh at Noon	45	55	66	68	74	86	90	86	72	51	46	43	
All-Sky Solar Radiation	RadAvg	1706	1937	2148	2249	2370	2358	2168	2148	2198	2094	1802	1666	
	RadStd	75	72	86	105	82	75	77	77	40	48	59	78	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	+0.59	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	+313	+306
Regional (2 neighbors)	+0.83	N/A	N/A	+1.06	N/A	N/A	N/A	-83	+338	+250

Nomenclature: See separate page